

HR Analytics — Employee Attrition Analysis

Executive Summary

Project Overview

This project analyzes employee workforce data to identify patterns and key factors associated with employee attrition. The analysis was conducted using **MySQL** for data validation, exploratory data analysis, business-question analysis, and advanced SQL techniques, followed by **Microsoft Power BI** for interactive visualization and business intelligence reporting.

The dataset contains **1,470 employees across 35 attributes**, covering employee demographics, organizational structure, compensation, job satisfaction, work environment, career progression, overtime, business travel, and attrition status.

The primary objective was to transform employee-level data into actionable insights that can support **data-driven workforce management and employee retention strategies**.

Executive Workforce Snapshot

Metric	Result
Total Employees	1,470
Employees Who Left	237
Active Employees	1,233
Overall Attrition Rate	16.12%
Average Employee Age	36.9 years
Average Monthly Income	6,502.93

The organization has an overall attrition rate of **16.12%**, meaning approximately 16 out of every 100 employees in the dataset have left the organization.

Major Findings

1. Early-Career Employees Represent a Major Retention Risk

Employees aged **18–25** recorded the highest age-group attrition rate at **35.77%**.

Employees with **0–2 years of tenure** also experienced a significantly higher attrition rate of **29.82%**.

This indicates that the organization may need stronger early-career onboarding, mentorship, career development, and employee engagement initiatives.

2. Sales Shows the Highest Departmental Attrition

The **Sales department** recorded the highest departmental attrition rate at **20.63%**, compared with:

- Human Resources — **19.05%**
- Research & Development — **13.84%**

The **Sales Representative** role recorded the highest job-role attrition rate at **39.76%**.

This makes the Sales workforce an important area for further retention investigation.

3. Compensation Is Strongly Associated With Attrition

Employees earning **up to 5K** recorded an attrition rate of **21.76%**, while employees earning **15K+** recorded only **3.76%**.

Employees at **Job Level 1** also experienced the highest job-level attrition rate at **26.34%**.

These findings highlight the importance of reviewing entry-level compensation, career progression, and reward structures.

4. Overtime Is Associated With Higher Attrition

Employees working overtime recorded an attrition rate of **30.53%**, compared with **10.44%** among employees who did not work overtime.

The substantial difference suggests that workload, working hours, and employee well-being should be investigated as potential retention factors.

5. Employee Satisfaction Matters

Employees with the lowest:

- **Job Satisfaction** recorded **22.84%** attrition.
- **Environment Satisfaction** recorded **25.35%** attrition.
- **Work-Life Balance** recorded **31.25%** attrition.

These results indicate that employee experience and workplace conditions should be considered important components of the organization's retention strategy.

6. Frequent Business Travel Is Associated With Higher Attrition

Employees who **travel frequently** recorded an attrition rate of **24.91%**, compared with **8.00%** among employees who do not travel.

This suggests that frequent travelers may require additional support, workload management, or travel-related interventions.

Business Significance

The analysis reveals that employee attrition is **not evenly distributed across the workforce**.

The highest-risk groups include:

- Younger employees
- Early-tenure employees
- Job Level 1 employees
- Lower-income employees
- Sales Representatives
- Employees working overtime
- Employees with low satisfaction levels
- Employees with poor work-life balance
- Frequent business travelers

These patterns provide HR management with specific workforce segments that can be prioritized for retention initiatives rather than applying a single strategy across the entire organization.

Recommended Management Priorities

Based on the analysis, management should prioritize:

1. Early-Career Retention

Strengthen onboarding, mentorship, career development, and progression opportunities for new and junior employees.

2. Sales Workforce Retention

Investigate workload, compensation, incentives, career progression, and management practices within Sales, particularly among Sales Representatives.

3. Compensation Review

Evaluate compensation competitiveness and progression opportunities for lower-paid and junior employees.

4. Overtime & Workload Management

Monitor overtime exposure and investigate workload-related employee burnout.

5. Employee Experience

Strengthen employee feedback mechanisms and address factors associated with low job and environment satisfaction.

6. Work-Life Balance

Review workload allocation, scheduling, flexibility, and employee well-being initiatives.

7. Continuous Attrition Monitoring

Establish a recurring HR analytics process to monitor attrition across department, job role, tenure, salary band, job level, satisfaction, overtime, and business travel.

Analytical Conclusion

The analysis demonstrates how HR data can be transformed into actionable workforce intelligence.

With an overall attrition rate of **16.12%**, the organization has identifiable employee segments experiencing substantially higher turnover than the company average. The strongest signals appear among **early-career employees, junior job levels, lower-income employees, Sales Representatives, overtime workers, and employees reporting lower satisfaction or work-life balance.**

Rather than treating attrition as a single organization-wide problem, management can use these insights to develop **targeted retention strategies based on employee segment and workplace characteristics.**

The accompanying Power BI dashboard provides an interactive platform for monitoring these workforce patterns and supporting evidence-based HR decision-making.

Analytical Note

Important: The findings in this analysis identify associations between employee characteristics and attrition. They should not automatically be interpreted as proof of causation. Further qualitative investigation, employee feedback, and longitudinal analysis would be required to establish causal relationships.

Tools Used

MySQL | SQL | Microsoft Power BI | DAX | Excel

Project Deliverables

- **SQL Analysis** — Data validation, EDA, attrition analysis, and advanced SQL.
- **Power BI Dashboard** — Five-page interactive HR analytics dashboard.
- **Executive Summary** — High-level findings and business significance.
- **Business Problem & Analytical Objective** — Analytical framework and business questions.
- **Insights & Recommendations** — Detailed findings and recommended HR actions.